

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

Lecture 1-2: What is Machine Learning?

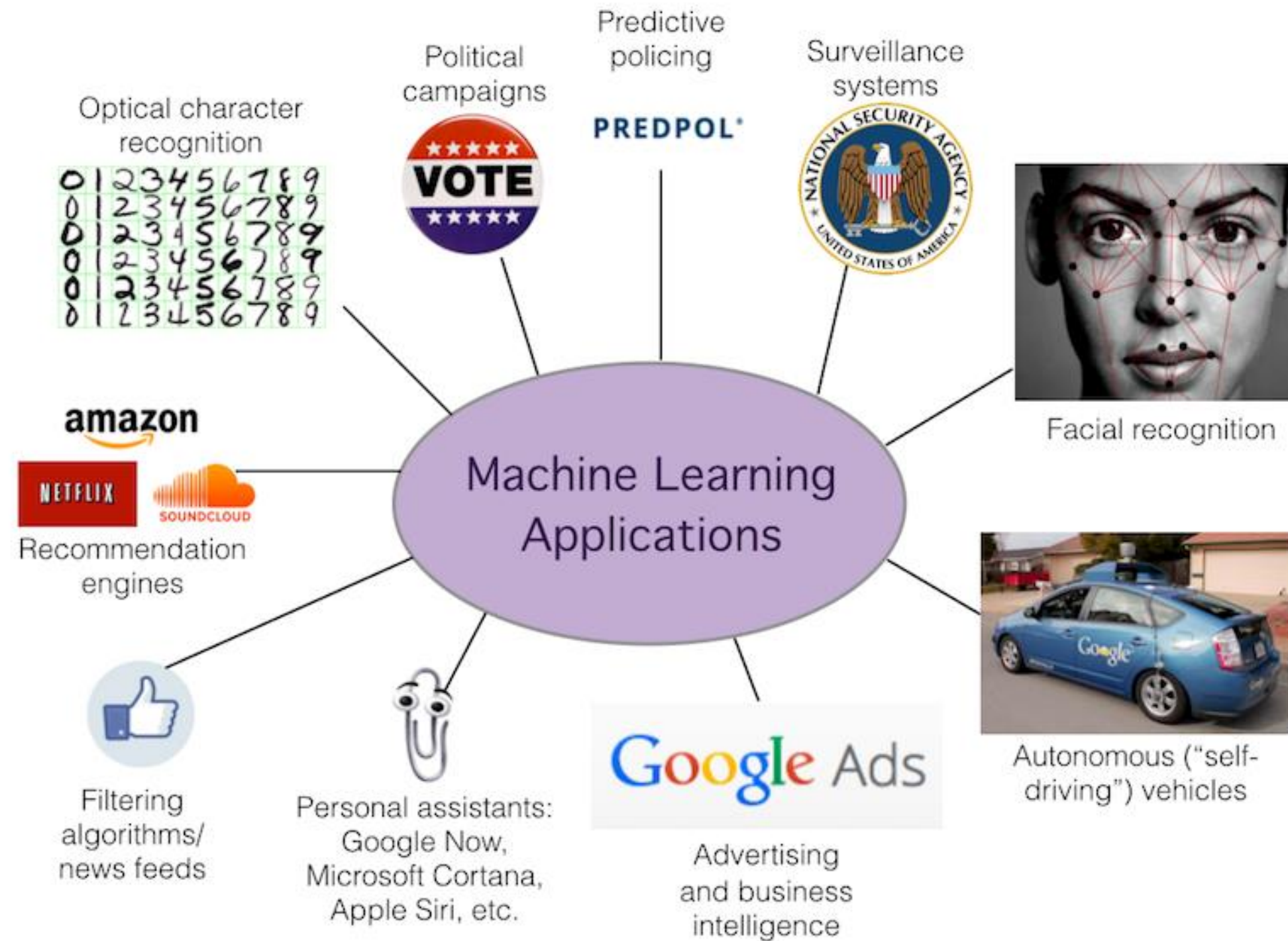
Texas A&M University

Bobak Mortazavi

Topics

- Introduction to Machine Learning
 - What is machine learning?
 - Models and accuracy
 - Some general supervised learning examples

What is machine learning?



Let's start with an example

- Let's say you want to teach a computer how to tell the difference between a cat and dog.



How do you describe the differences?

- Let's say you want to teach a computer how to tell the difference between a cat and dog



- Assume you take these pictures (call this **data collection**)
- Take a moment and ask yourself – how would you tell the difference between a cat and dog?

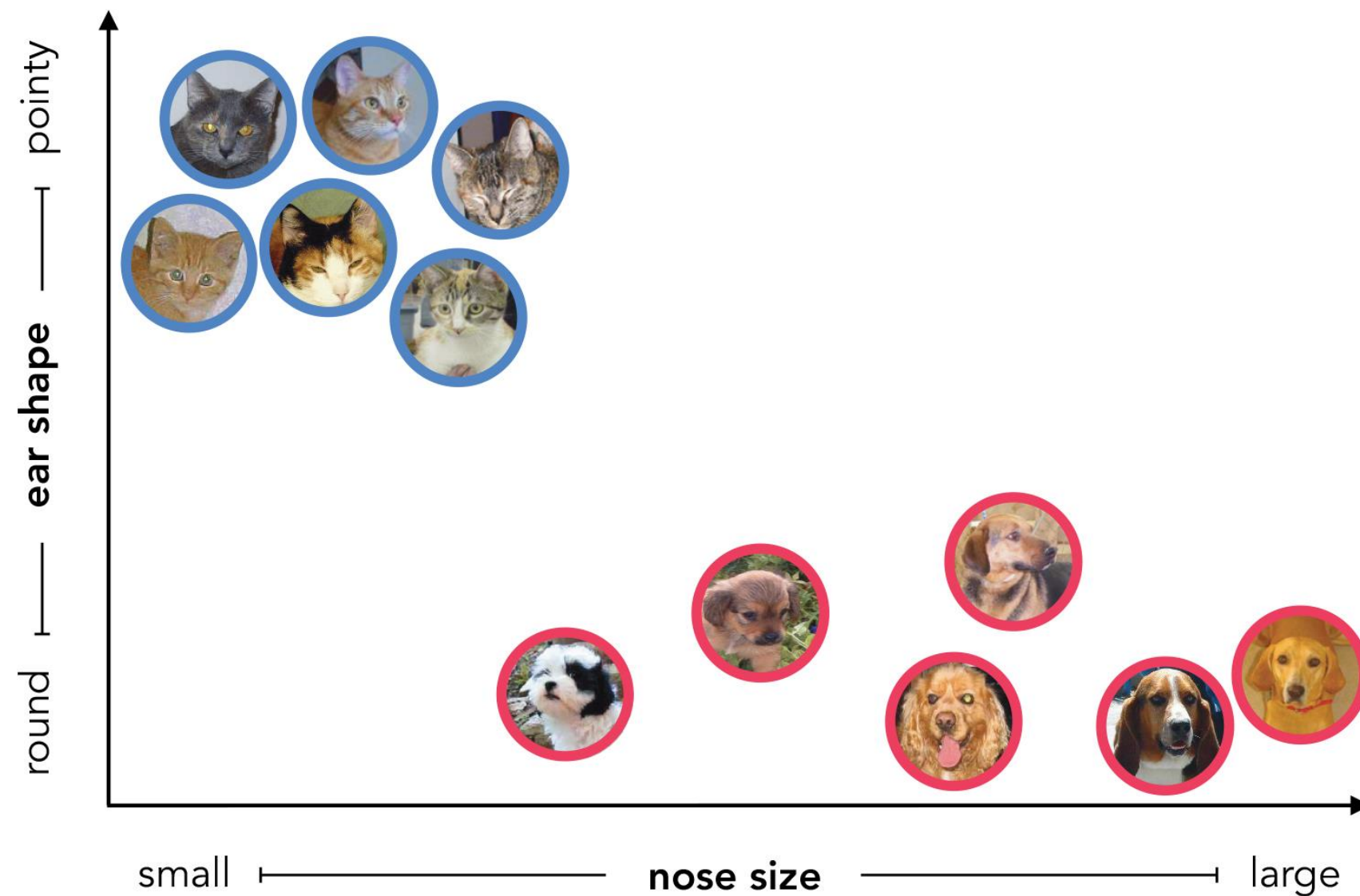
Feature extraction

- Let's say you want to teach a computer how to tell the difference between a cat and dog.



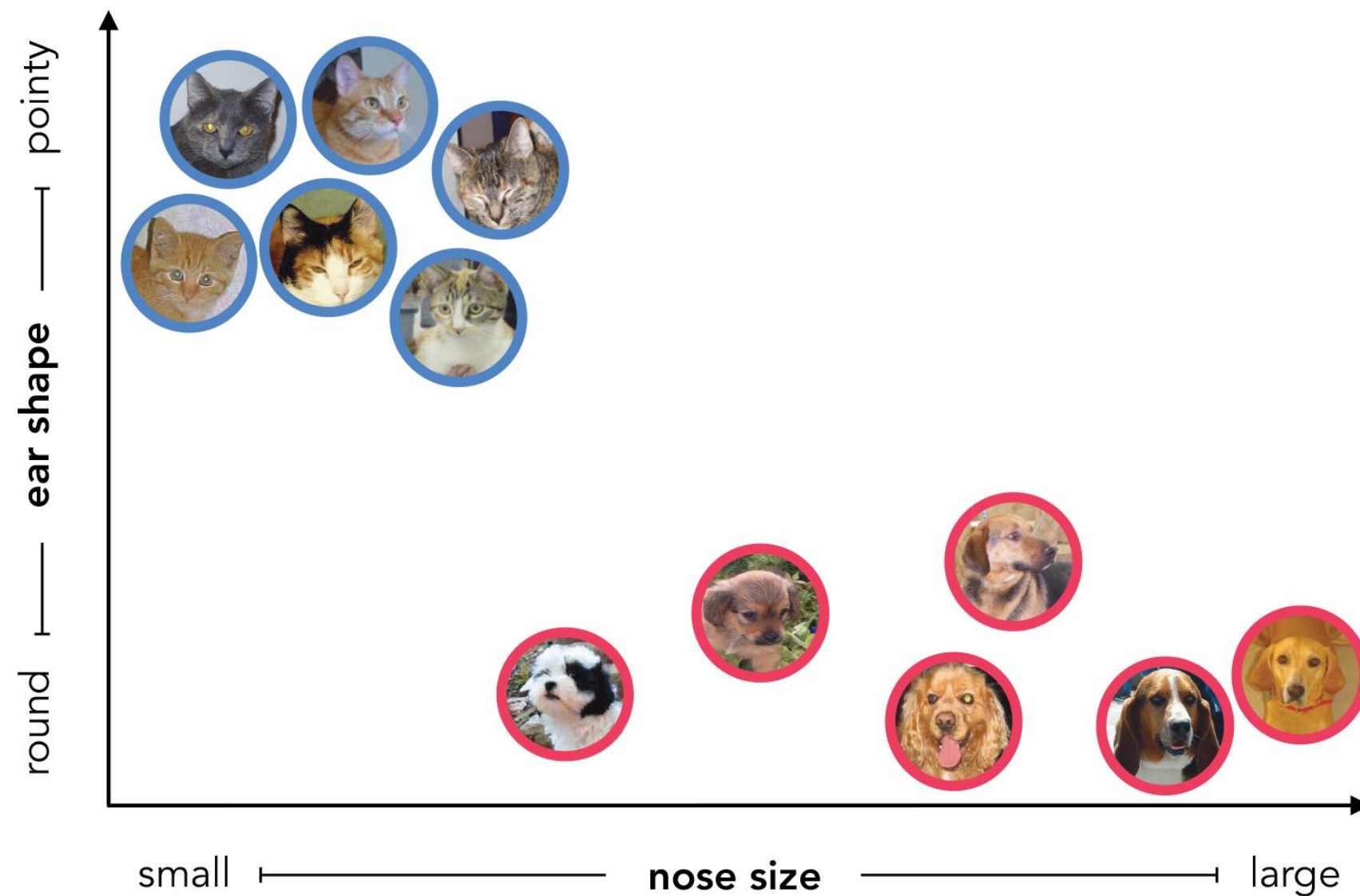
- Those things you just described? Those can be considered **features** (or **covariates**)
- So now we have **data collection** and **feature design** (or **feature engineering**) as two important steps.

Picking good features! (1)



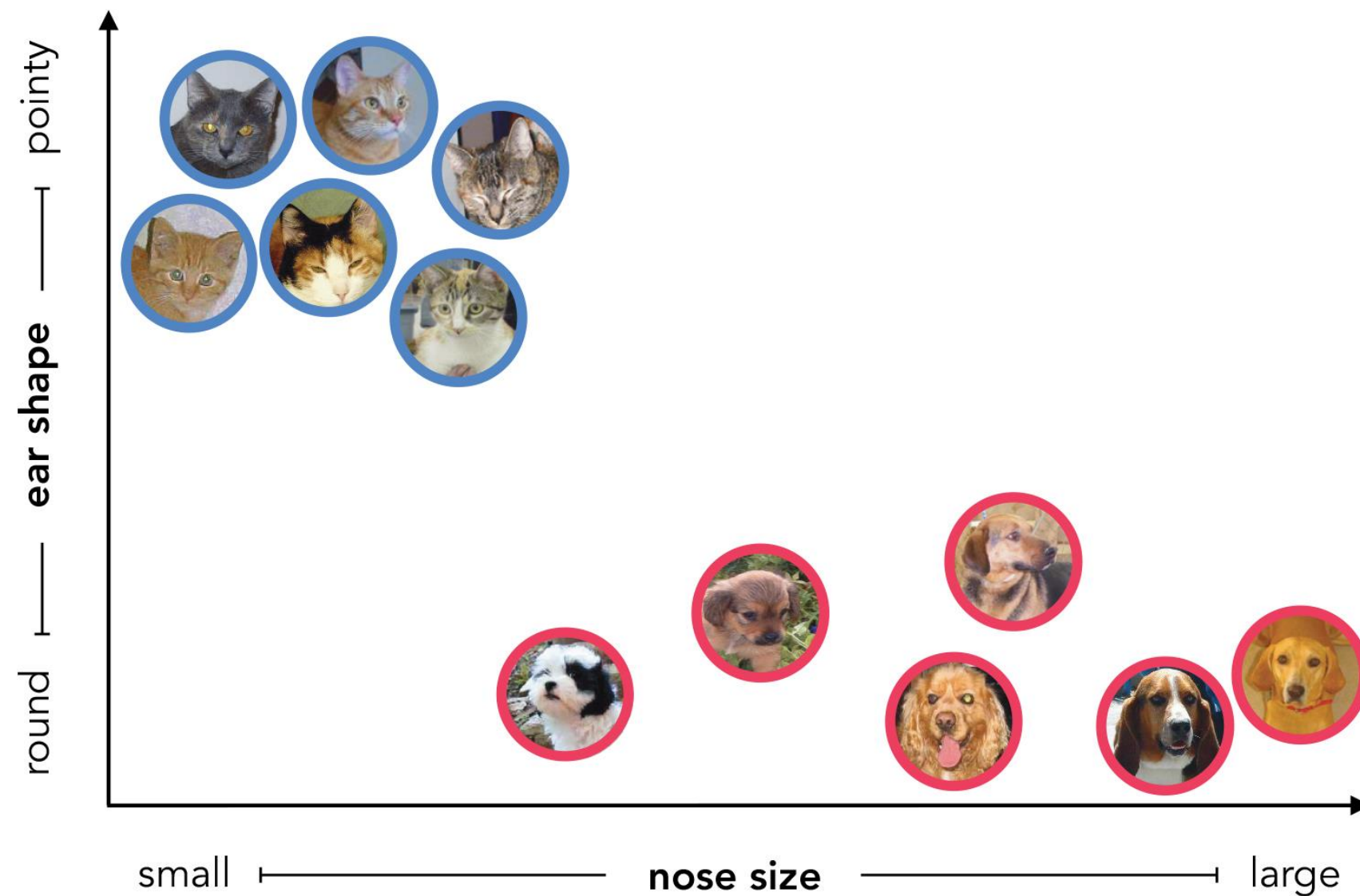
- Let's see how well some features distinguish
- Example: *size of nose* and *shape of ear*

Picking good features! (2)



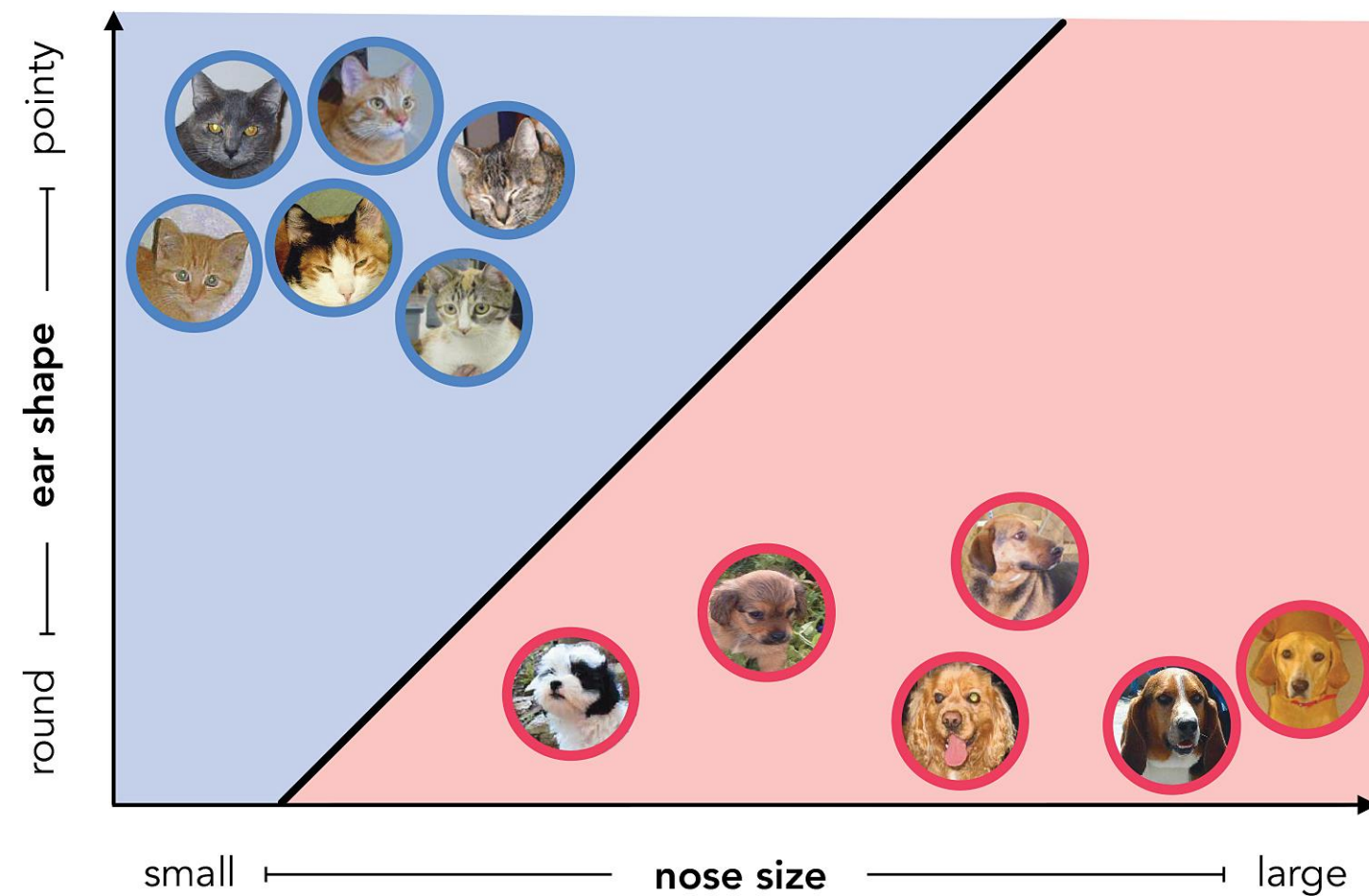
- Let's see how well some features distinguish
- Example: *size of nose* and *shape of ear*
- These seem to separate (or discriminate) between cats and dogs well!

Picking good features! (3)



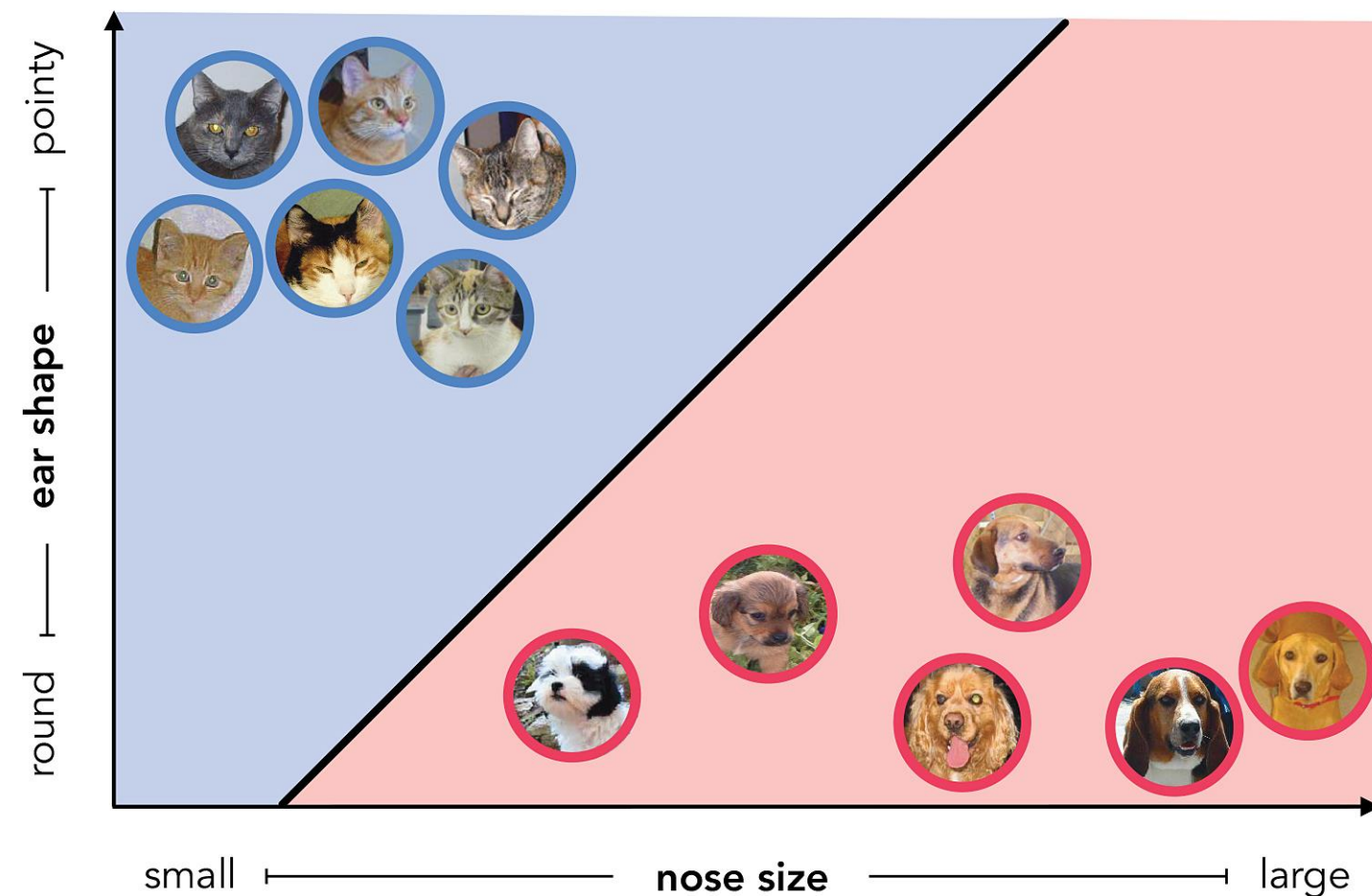
- Let's see how well some features distinguish
- Example: *size of nose* and *shape of ear*
- These seem to separate (or discriminate) between cats and dogs well!
- Can we mathematically describe this difference?

Designing a Model (1)



- This line seems to separate the space where cats and dogs are represented
- How would you describe this line that separates these objects?

Designing a Model (2)



- This line seems to separate the space where cats and dogs are represented
- The line is described by a Slope + Intercept. Finding these terms is done via different mathematical optimization techniques.

Testing your model (1)

- Are these dogs or cats?

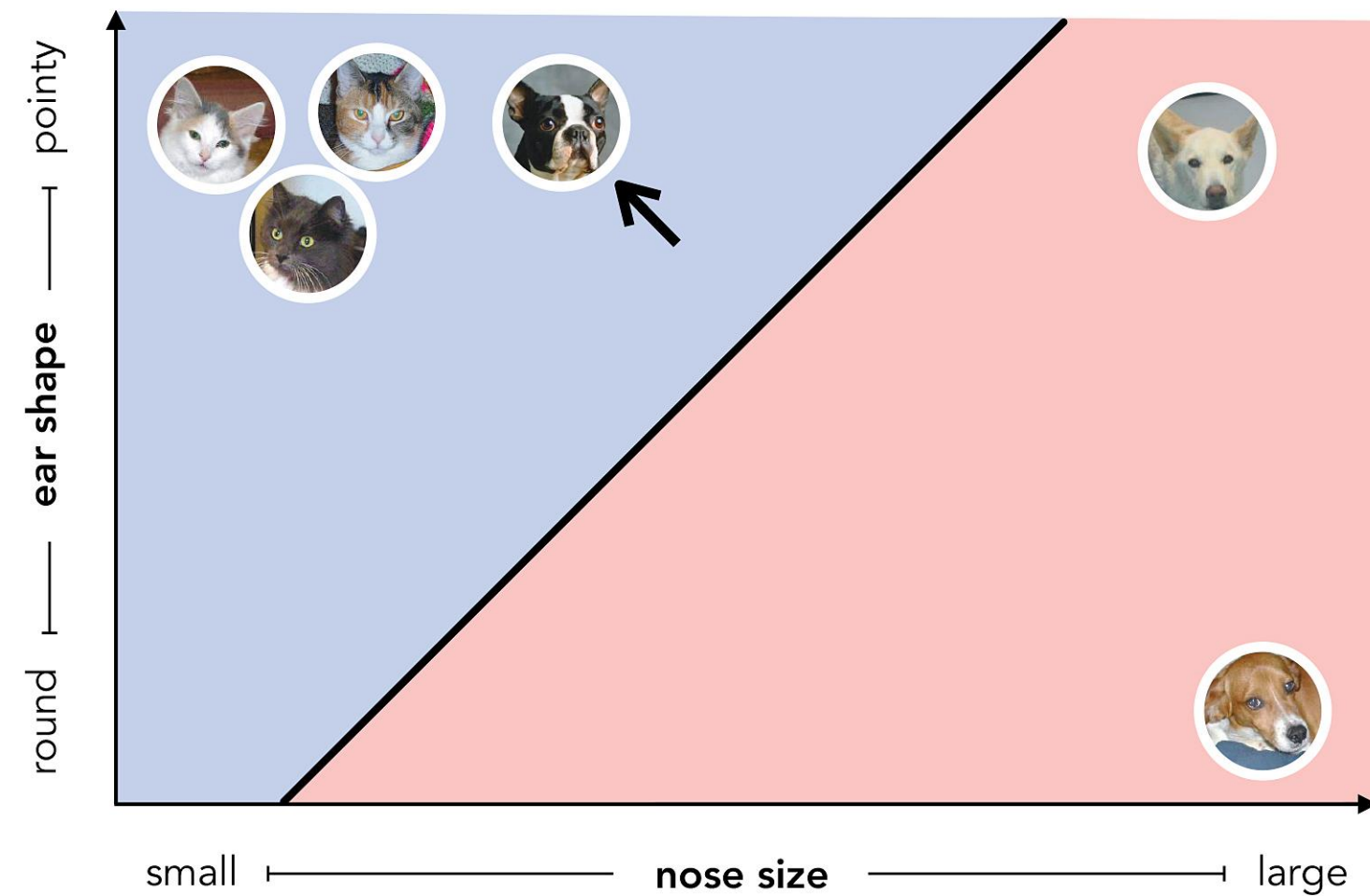


Testing your model (2)



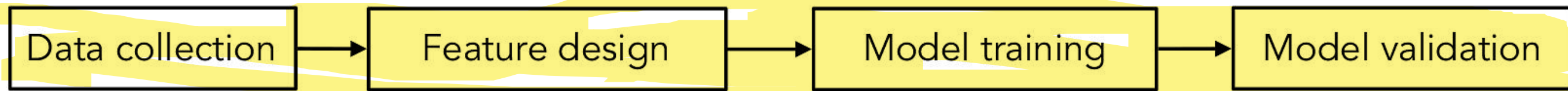
- Are these dogs or cats?
- How would you validate your model?
- What ways would you describe if your model is good?

Evaluating Test Cases



- After you build a model, you must validate that it works on new, unseen data!
- Might not work as well as the training version

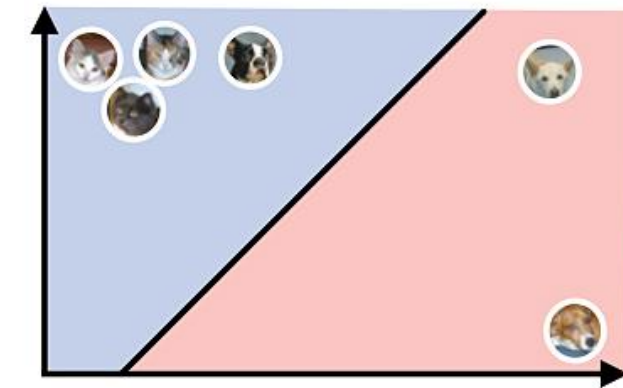
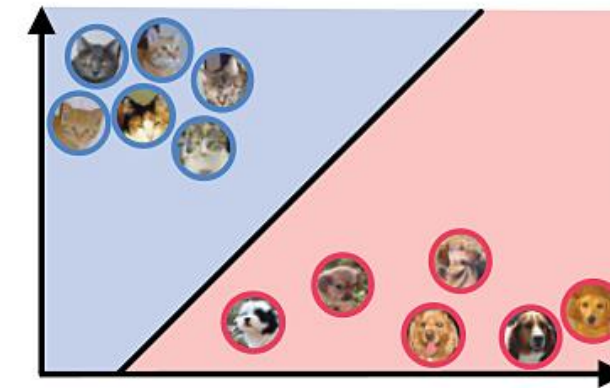
General Machine Learning Pipeline



Training set



Validation set



Takeaways

- Introduction to Machine Learning
 - What is machine learning?
 - Learned what are ML models
 - Learned how a model is trained and Tested